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!pip install requests beautifulsoup4 pandas
import requests
import time # Import the time module

url = "https://books.toscrape.com/catalogue/page-1.html"
response = requests.get(url)

print(response.status_code) # 200 means success
from bs4 import BeautifulSoup

soup = BeautifulSoup(response.text, 'html.parser')
books = soup.find_all('article', class_='product_pod') # each book is inside this tag

titles = []
prices = []

for book in books:
    title = book.h3.a['title'] # get title from 'title' attribute
    price = book.find('p', class_='price_color').text # get price text

    titles.append(title)
    prices.append(price)
    import pandas as pd

data = {'Book Title': titles, 'Price': prices}
df = pd.DataFrame(data)

df.to_csv('books_dataset.csv', index=False)

print("Done! File saved as books_dataset.csv")
print(df.head()) # shows first 5 rows
all_titles = []
all_prices = []

for page in range(1, 3): # scrape page 1 and 2 only
    url = f"https://books.toscrape.com/catalogue/page-{page}.html"
    response = requests.get(url)
    soup = BeautifulSoup(response.text, 'html.parser')
    books = soup.find_all('article', class_='product_pod')

    for book in books:
        all_titles.append(book.h3.a['title'])
        all_prices.append(book.find('p', class_='price_color').text)

    time.sleep(2) # wait 2 seconds so we don't overload the site

df = pd.DataFrame({'Book Title': all_titles, 'Price': all_prices})
df.to_csv('books_dataset_full.csv', index=False)

```

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Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (2.32.4)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.12/dist-packages (4.13.5)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests) (3.4.7)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests) (3.13)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests) (2026.4.22)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4) (2.8.3)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4) (4.15.0)
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17.0)
200
Done! File saved as books_dataset.csv

```

	Book Title	Price
0	A Light in the Attic	£51.77
1	Tipping the Velvet	£53.74
2	Soumission	£50.10
3	Sharp Objects	£47.82
4	Sapiens: A Brief History of Humankind	£54.23

```

import pandas as pd
import matplotlib.pyplot as plt

# 1. Load data

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df = pd.read_csv('books_dataset.csv')

# 2. Clean price column
# The 'Â£' symbol is often a result of encoding issues, replace it correctly
df['Price'] = df['Price'].str.replace('Â£', '').astype(float)

# 3. Basic info
print("First 5 rows:\n", df.head())
print("\nData info:\n", df.info())
print("\nStatistics:\n", df.describe())

# 4. Check missing and duplicates
print("\nMissing values:\n", df.isnull().sum())
print("\nDuplicates:", df.duplicated().sum())

# 5. Visualization
plt.hist(df['Price'], bins=20, color='skyblue', edgecolor='black')
plt.title('Book Price Distribution')
plt.xlabel('Price (£)')
plt.ylabel('Count')
plt.show()
```

First 5 rows:

	Book Title	Price
0	A Light in the Attic	51.77
1	Tipping the Velvet	53.74
2	Soumission	50.10
3	Sharp Objects	47.82
4	Sapiens: A Brief History of Humankind	54.23

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 20 entries, 0 to 19

Data columns (total 2 columns):

#	Column	Non-Null Count	Dtype
0	Book Title	20 non-null	object
1	Price	20 non-null	float64

dtypes: float64(1), object(1)

memory usage: 452.0+ bytes

Data info:

None

Statistics:

	Price
count	20.000000
mean	38.048500
std	15.135231
min	13.990000
25%	22.637500
50%	41.380000
75%	51.865000
max	57.250000

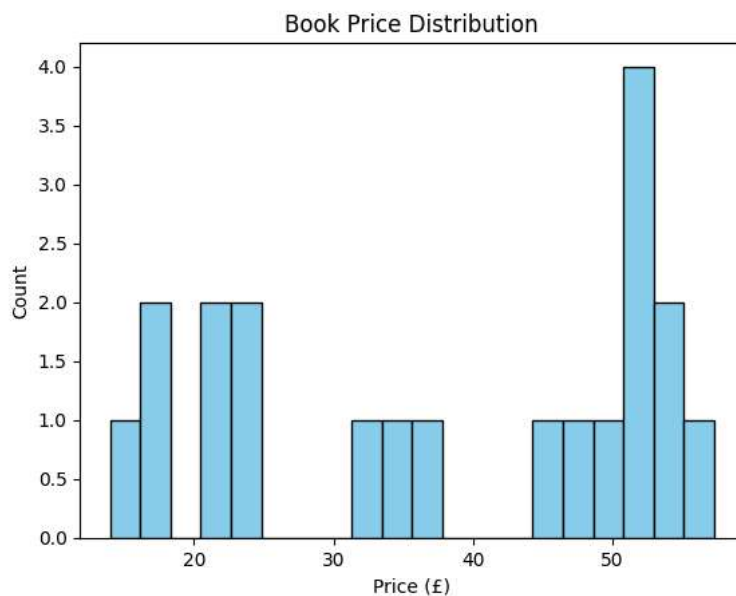
Missing values:

Book Title 0

Price 0

dtype: int64

Duplicates: 0



```
import pandas as pd
import matplotlib.pyplot as plt

# Load the CSV file
df = pd.read_csv('books_dataset.csv')

# First 5 rows - what does the data look like?
print(df.head())

# How many rows and columns?
print("Shape:", df.shape)
# Show column names and data types
print(df.info())

# Show basic statistics for number columns
print(df.describe())
df['Price'] = df['Price'].str.replace('Â£', '').astype(float)
# Most expensive and cheapest book
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```
# Most expensive and cheapest book
print("Most expensive:", df['Price'].max())
print("Cheapest:", df['Price'].min())

# Average price
print("Average price:", df['Price'].mean())

# Check for missing values
print("Missing values:\n", df.isnull().sum())
# Histogram of prices
plt.hist(df['Price'], bins=20, color='skyblue', edgecolor='black')
plt.title('Distribution of Book Prices')
plt.xlabel('Price (£)')
plt.ylabel('Number of Books')
plt.show()
cheap_books = df[df['Price'] < 30]
print(f"Books under £30: {len(cheap_books)} out of {len(df)}")
# Check for duplicate book titles
duplicates = df[df.duplicated(subset=['Book Title'])]
print("Duplicate books:", len(duplicates))

# Check for outliers - books much more expensive than others
outliers = df[df['Price'] > df['Price'].quantile(0.95)]
print("Possible outliers:\n", outliers[['Book Title', 'Price']].head())
```

	Book Title	Price
0	A Light in the Attic	£51.77
1	Lipping the Velvet	£53.74

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load and clean
df = pd.read_csv('books_dataset.csv')
df['Price'] = df['Price'].str.replace('£', '').astype(float)
df['Price Range'] = pd.cut(df['Price'], bins=[0, 20, 40, 60, 100],
                           labels=['Cheap 0-20', 'Medium 20-40', 'Expensive 40-60', 'Very Expensive 60+'])

# Dashboard with 3 charts
fig, axes = plt.subplots(1, 3, figsize=(15,4))

sns.histplot(df['Price'], bins=20, ax=axes[0], color='skyblue', kde=True)
axes[0].set_title('Price Distribution')

sns.countplot(data=df, x='Price Range', ax=axes[1], palette='pastel')
```